

Practice problems 5 (supplement on the Markov property)

Probability Theory (M. Math.)

1. Let S be a complete separable metric space with a sigma algebra $\mathcal{S}$. Let $(P_x)_{x \in S}$ be a family of probabilities in $(S, \mathcal{S})$ such that for any $A \in \mathcal{S}$, $x \mapsto P_x(A)$ is $\mathcal{S}$ -measurable. Define $P_x^{(1)}(A) := P_x(A)$, $P_x^{(2)}(A) := \int_S P_{x_1}(A) dP_x(x_1)$, and for $k > 2$,

$$\begin{aligned} P_x^{(k)}(A) &:= \int_{S^{k-1}} \int_A dP_{x_{k-1}}(x_k) \dots dP_x(x_1) \\ &= \int_{S^{k-1}} P_{x_{k-1}}(A) dP_{x_{k-2}}(x_{k-1}) \dots dP_x(x_1) \\ &= \int_S P_{x_1}^{(k-1)}(A) dP_x(x_1). \end{aligned}$$

Let λ be an $\mathcal{S}$ -probability in S . Define for $i_1 < \dots < i_k$, probabilities $\mu_{i_1, \dots, i_k}$ in S^{k+1} , such that for $A_0, A_1, \dots, A_k \in \mathcal{S}$,

$$\begin{aligned} &\mu_{i_1, \dots, i_k}(A_0, A_1, \dots, A_k) \\ &= \int_{S^{k+1}} \mathbb{1}_{A_0}(x) \mathbb{1}_{A_1}(x_1) \dots \mathbb{1}_{A_k}(x_k) dP_{x_{k-1}}^{(i_k - i_{k-1})}(x_k) \dots dP_x^{(i_1)}(x_1) d\lambda(x). \end{aligned}$$

- (a) Show that $(\mu_{i_1, \dots, i_k})_{i_1, \dots, i_k}$ is consistent, that is, for $i_1 < \dots < i_k < i_{k+1}$,

$$\mu_{i_1, \dots, i_{k+1}}(A_0, A_1, \dots, A_k, S) = \mu_{i_1, \dots, i_k}(A_0, A_1, \dots, A_k),$$

where $A_j \in \mathcal{S}$.

- (b) Use Kolmogorov's extension theorem to get a stochastic process $Z_0, Z_1, \dots$ and a probability $\mathbb{P}_\lambda$ in $S^{\mathbb{N}_0}$ such that

$$\mathbb{P}_\lambda[Z_0 \in A_0, Z_{i_1} \in A_{i_1}, \dots, Z_{i_k} \in A_{i_k}] = \mu_{i_1, \dots, i_k}(A_0, A_1, \dots, A_k),$$

for any choice $i_1 < \dots < i_k$ and $A_j \in \mathcal{S}$, which is in other words,

$$(Z_0, Z_{i_1}, \dots, Z_{i_k})_* \mathbb{P}_\lambda = \mu_{i_1, \dots, i_k}.$$

- (c) Verify in particular that $(Z_k)_* \mathbb{P}_\lambda = \lambda * P_{(\cdot)}^{(k)}$, that is

$$(Z_k)_* \mathbb{P}_\lambda(A) = \int_S P_x^{(k)}(A) d\lambda(x),$$

for any $A \in \mathcal{S}$. (Use the fact $\mathbb{P}_\lambda[Z_k \in A] = \mu_k(S, A)$.)

(d) Show that for bounded $\mathcal{S}$ -measurable $\psi : S \rightarrow \mathbb{R}$,

$$\begin{aligned} & \mathbb{E}_{\mathbb{P}_\lambda}[\psi \circ Z_k \cdot \mathbb{1}_{A_k} \circ Z_k \cdots \mathbb{1}_{A_0} \circ Z_0] \\ &= \int_{S^{k+1}} \psi(x_n) \mathbb{1}_{A_n}(x_n) \cdots \mathbb{1}_{A_0}(x_0) d\lambda(x) dP_x(x_1) \cdots dP_{x_{n-1}}(x_n). \end{aligned}$$

(Check above for $\psi = \mathbb{1}_B$, by noting that $(Z_0, \dots, Z_n)_* \mathbb{P}_\lambda = \mu_{1, \dots, n}$ by definition, and then use MClassT.)

2. (Markov property) Recall the shift map $T : S^{\mathbb{N}_0} \rightarrow S^{\mathbb{N}_0}$, defined as

$$Z_i \circ T^j = Z_{i+j},$$

for all $i, j \in \mathbb{N}_0$. Show that for any $\mathcal{F}_\infty = \sigma(Z_0, Z_1, \dots)$, measurable $\varphi : S^{\mathbb{N}_0} \rightarrow \mathbb{R}$,

$$\mathbb{E}_{\mathbb{P}_\lambda}[\varphi \circ T^n | \mathcal{F}_n](\omega) = \mathbb{E}_{Z_n(\omega)}[\varphi],$$

$\mathbb{P}_\lambda$ -a.e ω by checking the following. (Here we use the notational convention $\mathbb{E}_x[\cdot] = \mathbb{E}_{\mathbb{P}_x}[\cdot]$.)

(a) Check for $\varphi(\omega) = \mathbb{1}_{B_0} \circ Z_0(\omega) \cdots \mathbb{1}_{B_m} \circ Z_m(\omega)$ that for any $\mathbb{1}_{\{Z_0 \in A_0, \dots, Z_n \in A_n\}}$ (which is $\mathcal{F}_n$ -measurable), that

$$\begin{aligned} & \int_{S^{\mathbb{N}_0}} \mathbb{E}_{\mathbb{P}_\lambda}[\varphi \circ T^n | \mathcal{F}_n](\omega) \mathbb{1}_{\{Z_0 \in A_0, \dots, Z_n \in A_n\}}(\omega) d\mathbb{P}_\lambda(\omega) \\ & \stackrel{a1}{=} \int (\mathbb{1}_{B_0} \circ Z_n \cdots \mathbb{1}_{B_m} \circ Z_{m+n}) \cdot (\mathbb{1}_{A_0} \circ Z_0 \cdots \mathbb{1}_{A_n} \circ Z_n) d\mathbb{P}_\lambda(\omega) \\ & \stackrel{a2}{=} \int_{S_{n+m}} \mathbb{1}_{A_0}(x_0) \cdots \mathbb{1}_{A_n \cap B_0}(x_n) \mathbb{1}_{B_1}(x_{n+1}) \cdots \mathbb{1}_{B_m}(x_{n+m}) d\lambda(x) dP_x(x_1) \cdots dP_{x_{n+m-1}}(x_{n+m}) \\ &= \int_{S_n} \mathbb{1}_{A_0}(x_0) \cdots \mathbb{1}_{A_n}(x_n) \psi(x_n) d\lambda(x_0) dP_{x_{n-1}}(x_n) \cdots dP_{x_0}(x_1), \end{aligned}$$

where

$$\begin{aligned} \psi(x_n) &:= \int_{S_m} \mathbb{1}_{B_0}(y_0) \cdots \mathbb{1}_{B_m}(y_m) \delta_{x_n}(y_0) dP_{y_0}(y_1) \cdots dP_{y_{m-1}}(y_m) \\ & \stackrel{a3}{=} \mathbb{P}_{\delta_{x_n}}[Z_0 \in B_0, \dots, Z_m \in B_m] = \mathbb{E}_{x_n}[\varphi]. \end{aligned}$$

Argue using definition of CE for (a1) and using Problem 1(b) for (a2) and (a3).

(b) By Problem 1(d) and 2(a) conclude that

$$\begin{aligned} & \int_{S^{\mathbb{N}_0}} \mathbb{E}_{\mathbb{P}_\lambda}[\varphi \circ T^n | \mathcal{F}_n](\omega) \mathbb{1}_{\{Z_0 \in A_0, \dots, Z_n \in A_n\}}(\omega) d\mathbb{P}_\lambda(\omega) \\ &= \int_{S^{\mathbb{N}_0}} \psi \circ Z_n \cdot \mathbb{1}_{A_n} \circ Z_n \cdots \mathbb{1}_{A_0} \circ Z_0 d\mathbb{P}_\lambda(\omega) \\ &= \int_{S^{\mathbb{N}_0}} \mathbb{E}_{Z_n}[\varphi] \cdot \mathbb{1}_{A_n} \circ Z_n \cdots \mathbb{1}_{A_0} \circ Z_0 d\mathbb{P}_\lambda(\omega). \end{aligned}$$

(c) Use MClassT to show that Problem 2(b) can be extended to bounded measurable φ .

(d) Conclude by showing the claim.

3. Show that for $j > i$

$$\mathbb{E}_{\mathbb{P}_\lambda}[\varphi \circ T^j | \mathcal{F}_i](\omega) = \mathbb{E}_{Z_i(\omega)}[\varphi \circ T^{j-i}],$$

$\mathbb{P}_\lambda$ -a.e ω .

4. Show that for $j > i$

$$\mathbb{E}_{\mathbb{P}_\lambda}[\mathbb{1}_A \circ Z_j | \mathcal{F}_i](\omega) = P_{Z_i(\omega)}^{(j-i)}(A) = \mathbb{E}_{\mathbb{P}_\lambda}[\mathbb{1}_A \circ Z_j | \sigma(Z_i)](\omega),$$

$\mathbb{P}_\lambda$ -a.e ω . (Apply Problem 2 to $\varphi = \mathbb{1}_A \circ Z_1$ and then use Problem 1(c)).